

VERSION CONTROL

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Go to Github Website and create an account https://github.com › signup

1. Github Profile

Create a new repository in your Github Profile

Setting up Git on Local Computer

- Install VS Code
- Install Git Bash for windows https://git-scm.com/downloads
- Terminal for MAC
- After installation on Git Bash and check version

```
git --version
```

Configuring Git

```
Git config –global user.name “My name”
Git config –global user.email  “Myname@email.com”
```

Check the account

```
git config –list
```

PART A – GitHub (Remote Repository) to VS Code (Local Repository)

1. Clone and Status

Clone: Cloning a repository on our local machine

```
git clone <-link->
```

Status: display the state of the code

```
git status
```

- Go to github → click on code and copy https link
- On VS code terminal paste the link in the following command
  `git clone https://....`

- You will see the folders from Github in VS code
- Make sure you are in the right directory
- Now check for the folders using `ls` and hidden folders using `ls -Force`. You will see multiple folders including `.git`

Making changes in local file and pushing it to github

- Change a file in VS
- Check `git status` (M - modified)
- To save the changes we have to use `add` and `commit`

Type of file status

- **Untracked:** created is VS but not in github or github doesn't track
- **Modified:** the file has been changed but not updated on github
- **Staged:** file is ready to be committed
- **Unmodified:** the file is unchanged

2. ADD AND COMMIT

Add: adds new or changed file in your working directory to git staging area

```
git add <-file name->
```

Commit: it is the record of change

```
git commit -m "some message"
```

Changes are added and committed but not reflected on github, for that you must use `push` command

3. PUSH

Push: uploads local repository content to remote repository

`git push origin main` (repository of git is named origin)

PART B – VS CODE (LOCAL REPOSITORY) TO GITHUB (REMOTE REPOSITORY)

`init` is used to create a new git repository

```
git init
```

```
git remote add origin <-link->
```

`git remote -v` (to verify remote)

`git branch` (to check branch)

`git branch -M main` (to rename branch)

In VScode create a new directory (local repository)

Make it a git repo first

```
git init
```

- check for hidden folder to make sure it contains `.git` folder
- check status

```
git add .
```

```
git commit -m "message"
```

```
git status
```

- Changes are made to the local repository, now add them to github
- Go to github profile and create a repository

- Go to VS terminal

  `git remote add origin <-link-> (https link from github)`

  `git remote -v`

  `git branch` (to check the branch)

- if the branch is `master,` rename it to `main`

  `git branch -M main`

  `git branch`

  `git push origin main`

  `git push origin main -u (to avoid repetition)`

Workflow

- Github Repo → Clone → Changes → Add → Commit → Push

1. Branch Commands

```
git branch (to check branch)
git branch -M main (to rename branch)
git checkout <-branch name-> (to navigate)
git checkout -b <-new branch name-> (to create new branch)
git branch -d <-branch name -> (to delete branch)
```

2. Merging code

WAY 1:

```
git diff <-branch name-> (to compare commit branch file and more)
git merge <-branch name-> (to merge 2 branches)
```

WAY 2:

- Create a PR (pull request)

```
Bring changes in VS after Pull/compare
Pull command

git pull origin main (used to fetch and download content from a
remote repo and immediately update the local repo to match
```

3. Resolving Merge Conflict

- PR
- GIT MERGE

4. UNDOING CHANGES

CASE 1: STAGED CHANGES

```
git reset <-filename>
git reset
```

CASE 2: COMMITTED CHANGES (FOR ONE COMMIT)

```
git reset HEAD~1
```

CASE 3: COMMITTED CHANGES (FOR MULTIPLE COMMITS)

```
git reset <-commit hash->
git reset –hard <-commit hash->
```

5. FORK

Rough copy – copying other github projects to your account